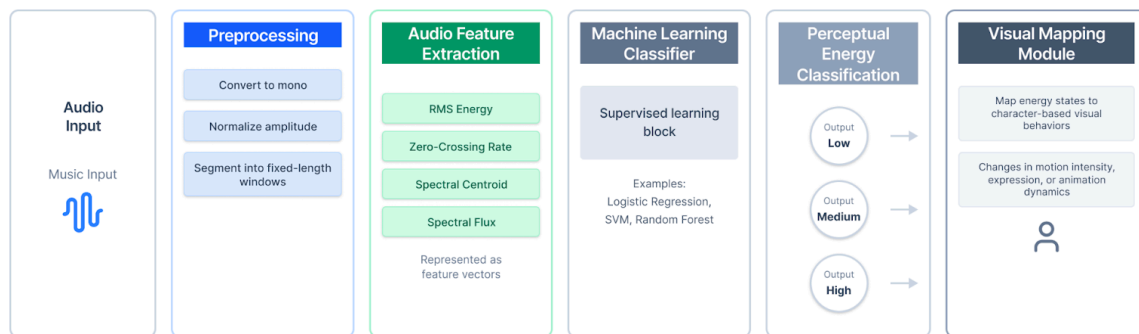


Team 18 Progress Update

Music-to-Imagine System

Goal

Build a simple pipeline that classifies music into **Low / Medium / High energy**, then maps it to **character visual behaviors**.



Luyao Niu at Sat Jan 24, 2026 10:26am

Be mindful of time needed for data collection

FMA: <https://www.tensorflow.org/datasets/catalog/gtzan>

- **Description:**

The dataset consists of 1000 audio tracks each 30 seconds long. It contains 10 genres, each represented by 100 tracks. The tracks are all 22050Hz Mono 16-bit audio files in .wav format.

The genres are:

- blues
- classical
- country
- disco
- hiphop
- jazz
- metal
- pop
- reggae
- rock

- **Additional Documentation:** [Explore on Papers With Code](#) ↗

- **Homepage:** <http://marsyas.info/index.html>

Current System Pipeline

Audio Input (44.1 kHz)

- **Preprocessing** (mono conversion, normalization, windowing 2048 samples, 50% overlap)
- **Feature Extraction** (RMS, ZCR, Spectral Centroid, Spectral Flux)
- **ML Classification** (baseline model: Logistic Regression / SVM / Random Forest)
- **Visual Mapping** (energy state controls motion speed + intensity)

DSP Features Used (Choose reasonable one method)

- **RMS Energy** (loudness/intensity)
- **Zero Crossing Rate** (noise/percussion vs tonal sound)
- **Spectral Centroid** (brightness)
- **Spectral Flux** (dynamic changes across frames)

